WEEK
10

Comprehensive Integration and Certification

Instruction | retrieval | multi-organ practice | field-ready reference

Public learner edition

This packet contains the teaching guide, cumulative reference, and formative practice for the week. Secure graded assessments remain in the online course so unfamiliar examination specimens and answer material are not published.

Student / participant

Date started

Planned pace / study notes

Field standard

Identify the narrowest rank the visible evidence can defend. A common name OR a scientific name satisfies an ordinary identity prompt unless the activity explicitly tests nomenclature. State uncertainty instead of converting weak evidence into a confident species claim.

Use this packet alongside the online course at cochetopa.co/course/. Work the instruction first, complete retrieval without looking back, and use the secure online assessment when ready for grading.

Week 10 overview

Learning objectives

1. Demonstrate rapid, calibrated identification of all eligible taxa from unfamiliar single-organ and mixed evidence.
2. Complete cumulative scientific-name retrieval in both directions.
3. Pass separate foliage, bark, reproductive, dormant, whole-tree/regional, and rapid mixed-evidence batteries.
4. Synthesize field silvics across site, regeneration, disturbance, succession, condition, and management.
5. Produce a final numerical grade and a separate species-by-modality mastery and certification report.

Species introduced this week

No new taxa. This week is comprehensive integration and certification.

Cumulative review

80 taxa remain eligible. Earlier species do not disappear from retrieval merely because the weekly theme changes.

Confusion sets

Sugar / black / Norway maple: Petiole latex; leaf underside hair and stipules; bud size/shape; samara angle
Request: Fresh-broken petiole; lower surface; terminal bud; ripe samara

Red / silver maple: Sinus depth; lower-surface whiteness; spring samara size; twig odor; bark strips
Request: Complete leaf underside; samara with scale; crushed twig; mature bark

Striped / mountain maple: Green-white striped bark versus unstriped; leaf size/base; fruit cluster posture; growth form
Request: Young stem; intact leaf; fruit cluster; whole plant

White / green / black ash: Leaf-scar shape and bud position; leaflet stalks; samara wing extent; black-ash corky bark
Request: Twig node and leaf scar; whole rachis; samara; bark; hydrologic setting

Paper / heart-leaved / gray birch: Leaf base and vein count; twig resin glands; peeling versus tight bark; black branch marks; elevation
Request: Several mature leaves; resin-warty twig; bark sheet; whole growth form; location

Yellow / sweet birch: Both wintergreen; yellow birch golden curling bark and spur shoots versus sweet birch dark blocky bark
Request: Scratched twig; young and mature bark; fruiting strobile; lower leaf surface

White / black / red / Norway spruce: Cone size and scale margin; twig hairs; bud resin; foliage color/odor; pendulous branchlets
Request: Full cone; twig macro; bud; whole crown; regional site

Spruce / fir / hemlock: Woody pegs versus pad base versus petiole; needle cross-section; cone orientation and persistence
Request: Needle attachment macro; underside; terminal shoot; intact or attached cone

Red / jack / Scots / pitch pine: Fascicle count/length; bend test; cone curvature/prickles; upper-bole orange bark; epicormic shoots
Request: Several complete fascicles; cone and umbo; upper and lower bark; whole crown

Tamarack / European larch: Cone size and scale count/edge; needle cluster density; native peatland versus planting
Request: Full cone; spur shoot; cluster count; site/planting context

Red-oak group: Bud hair/size; lower-leaf hair; sinus architecture; acorn cup depth/rings; bark; site and branch angle
Request: Several sun leaves; terminal buds; complete acorn/cup; mature bark; whole crown

White-oak group: Lower-leaf felt; fruit stalk; fringed cup; chestnut teeth; bark architecture; site
Request: Lower surface; acorn/cup/stalk; terminal bud; upper and lower bole; site

Aspens / balsam poplar / cottonwood: Petiole flattening; tooth size; leaf base; bud size/stickiness/odor; bark and floodplain/upland context
Request: Complete leaf and petiole; bud; young and old bark; clonal form; site

American / slippery / rock / Siberian elm: Leaf roughness/base; bud hair; corky wings; samara hair/notch; bark layering; crown
Request: Leaf surfaces; buds; samara; corky twigs; existing bark section; crown

Black / pin / chokecherry: Fruit cluster architecture; petiole glands; lower-midvein hair; bark age; whole pioneer/clonal form
Request: Raceme or cluster; petiole macro; lower leaf; mature bark; whole plant

American hornbeam / eastern hophornbeam: Muscular smooth bole and three-lobed fruit bracts versus shreddy bark and hoplike sacs
Request: Bole; fruit; bud; wet versus dry site

Black walnut / butternut: Terminal leaflet; rachis/twig hair; light versus dark pith; round versus elongated nut; bark
Request: Whole leaf; pith section; nut/husk; twig hair; mature bark

Bitternut / shagbark / pignut / mockernut hickory: Terminal bud color/size/hair; leaflet count; rachis hair; husk thickness/dehiscence; bark
Request: Terminal bud; entire leaf; husk/nut; mature bark; twig hair

Northern white-cedar / Atlantic white-cedar / eastern redcedar: Flattened fans versus fine false-cypress sprays versus rounded juniper twigs; cone type; fruit color; wetland chemistry
Request: Full spray; seed cone or berry-like cone; bark; site

Native compound leaves / invasive confusers: Opposite/alternate; bud position; leaflet glands; thorns; pith; leaf scar; odor; fruit type
Request: Intact leaf plus node; bud/leaf scar; pith; samara/pod/nut; leaflet base

Yellow birch / black cherry bark: Golden horizontal curls and wintergreen twig versus burnt-chip plates, petiole glands, and almond twig odor
Request: Young and old bark; scratched twig; leaf petiole and underside; fruit

Serviceberry / mountain-ashes / chokecherry: Simple versus compound leaves; pome calyx versus drupe; leaflet proportions; buds; fruit architecture
Request: Whole leaf; fruit close-up; bud; flowers; multiple leaves

Flowering / pagoda dogwood: Opposite versus alternate branching; red versus blue-black fruit; flower bud; crown tiers
Request: Intact twig nodes; fruit; bud; whole crown

White mulberry / sassafras / basswood: Latex and variable coarse leaves; sassafras aroma/mixed shapes; basswood asymmetric heart base and bracted fruit
Request: Fresh twig/petiole; multiple leaves; fruit; buds; bark

Assessment schedule

Assessment	Type	Items	Time
W10-NOMENCLATURE-FINAL	cumulative scientific name examination	80	60 min
W10-SILVICS-SYNTHESIS	oral or written silvics synthesis	40	100 min
W10-FIELD-CONDITION-CERT	field condition certification check	30	70 min
W10-FINAL-FOLIAGE	comprehensive final foliage only battery	20	35 min
W10-FINAL-BARK	comprehensive final bark only battery	20	40 min
W10-FINAL-REPRODUCTIVE	comprehensive final reproductive battery	16	30 min
W10-FINAL-DORMANT	comprehensive final dormant battery	20	45 min
W10-FINAL-WHOLE-TREE	comprehensive final whole tree regional uncertainty battery	20	40 min
W10-FINAL-RAPID-MIXED	comprehensive final rapid mixed evidence battery	50	70 min

Five-session field workflow

Session A - Rapid identification and remediation triage

rapid unfamiliar drills with no new taxa. Modalities: foliage, bark, reproductive, dormant, whole tree context, condition.

☐ Session A completed (enter X)

Session B - Cumulative scientific-name retrieval

closed-response nomenclature batches. Modalities: .

☐ Session B completed (enter X)

Session C - Confusers and insufficient evidence

isolated and side-by-side confuser drills with requested-view reasoning. Modalities: foliage, bark, reproductive, dormant, whole tree context, condition.

☐ Session C completed (enter X)

Session D - Certification modality rehearsal

separate foliage bark reproductive dormant and whole-tree stations. Modalities: foliage, bark, reproductive, dormant, whole tree context, condition.

☐ Session D completed (enter X)

Session E - Silvics and regional synthesis

oral or written synthesis and final targeted retrieval. Modalities: whole tree context, condition.

☐ Session E completed (enter X)

Evidence-to-decision protocol

Step	Field action
1. Inventory	Name only what is visible: organ, arrangement, attachment, surface, scale, maturity, damage, and context.
2. Generate	Name the leading identity and the nearest plausible alternative before seeking confirmation.
3. Separate	Cite the visible character that changes the decision; do not substitute habitat or color alone for structure.
4. Calibrate	Use species, genus, or insufficient evidence according to the photograph, not according to what you hope it shows.
5. Request	Ask for the single additional view with the highest expected diagnostic value.

Part I - Species instruction atlas

These profiles are labeled teaching material. Photographs come only from the teaching pool; no exact image in the formal assessment section appears here.

Part II - Cumulative rapid-reference atlas

Use this as a field briefing, not as a substitute for retrieval. Earlier taxa are compressed here so the packet remains self-contained while preserving space for unfamiliar assessment images.

Taxon	High-value visible evidence	Silvics anchor
Sugar maple *Acer saccharum*	Foliage: Opposite, simple blades are palmately veined and usually five-lobed, with pointed lobes, broad rounded U-shaped... Bark: Seedlings and saplings have smooth light-gray bark; pole trees develop long shallow furrows; mature and old... Dormant: Opposite brown twigs carry a narrow, sharply pointed terminal bud with many tightly overlapping...	Tolerance: Very shade tolerant; seedlings and saplings can persist for years as advance... Site: Best on cool, fertile, moist but well-drained loams with adequate calcium and magnesium; performs...
Red maple *Acer rubrum*	Foliage: Opposite, simple blades usually have three to five lobes, angular or V-shaped sinuses, and regular serration... Bark: Young stems are smooth and light gray; intermediate bark develops narrow vertical cracks and small scaly plates;... Dormant: Opposite reddish twigs bear blunt-ovoid red buds with several pairs of scales; flower buds are...	Tolerance: Moderately shade tolerant when young and able to persist beneath partial canopy,... Site: Occupies an exceptionally broad range from saturated swamps and floodplains to dry ridges; local seed...
American beech *Fagus grandifolia*	Foliage: Alternate, simple leaves are elliptic to ovate with a short petiole, straight parallel lateral veins, and one... Bark: Saplings and healthy mature trees have thin, smooth, pale bluish-gray bark, often compared with elephant hide.... Dormant: Alternate twigs carry exceptionally long, slender, sharply pointed cigar-shaped buds with many...	Tolerance: Exceptionally shade tolerant; seedlings, saplings, and root suckers can persist... Site: Most competitive on moist, well-drained, often fertile loams and slopes; absent from persistently...
Yellow birch *Betula alleghaniensis*	Foliage: Alternate ovate leaves have a rounded to weakly heart-shaped base, many straight lateral veins, and sharply... Bark: Seedlings are gray-brown and unremarkable; intermediate stems develop bronze to yellow-gold bark with horizontal... Dormant: Slender alternate twigs bear pointed reddish-brown buds, small spur shoots, and no true terminal...	Tolerance: Intermediate in shade tolerance: more tolerant than paper birch and aspen, but... Site: Best on cool, moist, well-drained, fertile loams and coves; common on north and east slopes, seepage...
Paper birch *Betula papyrifera*	Foliage: Alternate ovate to triangular leaves have a rounded or wedge-shaped base, pointed tip, and doubly serrate... Bark: Young stems are reddish-brown with pale horizontal lenticels and do not look white; pole and mature bark becomes... Dormant: Slender reddish-brown alternate twigs have pointed appressed buds, conspicuous resin glands or...	Tolerance: Very shade intolerant; seedlings require abundant light and are rapidly overtopped... Site: Occupies cool sites across a broad moisture range, from well-drained glacial tills and sandy soils to...
Eastern hemlock *Tsuga canadensis*	Foliage: Short, flat, mostly blunt single needles vary in length along delicate twigs, show two pale stomatal bands... Bark: Young bark is smooth gray-brown; intermediate stems develop narrow scaly ridges; mature bark becomes dark gray-... Dormant: Fine flexible yellow-brown twigs carry minute rounded buds, tiny petiolate needles, and often a...	Tolerance: Extremely shade tolerant, among the most tolerant North American trees; seedlings... Site: Favors cool, humid, sheltered sites with reliable moisture and generally well-drained acidic to...
American basswood *Tilia americana*	Foliage: Alternate, broadly ovate to heart-shaped leaves have an abruptly pointed tip, serrate margin, palmate basal... Bark: Young bark is smooth gray to gray-brown; intermediate bark develops narrow vertical fissures; old boles are dark... Dormant: Alternate zigzag twigs carry plump red to reddish-brown ovoid buds with only two or three...	Tolerance: Shade tolerant, though generally less persistent in deep shade than beech or sugar... Site: Best on deep, fertile, moist, well-drained loams with high base status, especially calcium-rich tills,...
Black cherry *Prunus serotina*	Foliage: Alternate, simple, elliptic leaves have fine rounded or incurved teeth, one or more small glands near the top of... Bark: Young stems are smooth reddish-brown to dark gray with prominent horizontal lenticels; intermediate bark breaks... Dormant: Slender alternate reddish-brown twigs carry small ovoid scaled buds and semicircular leaf scars;...	Tolerance: Shade intolerant; seedlings may survive briefly under partial canopy, but... Site: Best on deep, fertile, moist but well-drained loams, especially Allegheny uplands and protected...
Eastern hophornbeam *Ostrya virginiana*	Foliage: Alternate ovate leaves have many straight lateral veins, a sharply pointed tip, and a doubly serrate margin; the... Bark: Young stems are smooth gray-brown with lenticels; older small trunks break into narrow, loose, vertical gray-... Dormant: Slender alternate twigs bear short pointed ovoid buds with many visible scales; several...	Tolerance: Shade tolerant and able to persist as a slow-growing understory tree beneath... Site: Common on dry-mesic to mesic, well-drained, often rocky or calcareous soils; avoids prolonged flooding...
Black maple *Acer nigrum*	Foliage: Opposite blades are usually three-lobed, broad and somewhat drooping, with a deeply heart-shaped base, few... Bark: Young bark is gray and fairly smooth; older boles become dark gray-brown with irregular furrows and firm plates.... Dormant: Opposite brown twigs carry a pointed terminal bud and smaller paired laterals. Bud, twig, and...	Tolerance: Very shade tolerant when young, broadly like sugar maple; suppressed advance... Site: Best on fertile, moist, well-drained loams and alluvium, often over calcareous or otherwise base-rich...
Striped maple *Acer pensylvanicum*	Foliage: Opposite, large, thin blades have three shallow forward-pointing lobes, a broad heart-shaped base, and fine... Bark: Saplings have smooth green bark marked by conspicuous vertical white stripes; older stems turn greenish brown to... Dormant: Opposite green to green-brown twigs and young stems retain longitudinal pale stripes; paired...	Tolerance: Very shade tolerant; survives decades suppressed but grows best under moderate... Site: Favors cool, moist, well-drained sandy loams, commonly acidic, and avoids both stagnant wetness and...
Mountain maple *Acer spicatum*	Foliage: Opposite blades are usually three-lobed, sometimes weakly five-lobed, with sharply pointed lobes and coarse... Bark: Young stems are reddish to gray-brown without white longitudinal stripes; older small trunks remain thin-barked... Dormant: Slender opposite twigs carry small pointed paired buds and a terminal bud; stems lack striped...	Tolerance: Moderately shade tolerant and able to persist beneath canopy, but density and... Site: Favors cool, moist, well-drained upland soils, rocky slopes, ravines, and streamside benches; drought...
Norway maple *Acer platanoides*	Foliage: Opposite, broad blades usually have five sharply pointed lobes and sparse large teeth. A freshly broken petiole... Bark: Young bark is smooth gray; mature bark develops regular narrow vertical ridges and shallow interlacing furrows,... Dormant: Opposite stout twigs bear large plump terminal buds and paired lateral buds, commonly reddish...	Tolerance: Seedlings are shade tolerant to very shade tolerant and form persistent seedling... Site: Broadly site tolerant, performing especially well on cool moist rich soils while also tolerating urban...
Heart-leaved paper birch *Betula cordifolia*	Foliage: Alternate ovate leaves have a distinctly heart-shaped base, a pointed but not long-drawn tip, double teeth, and... Bark: Young stems are reddish to brown; mature bark becomes pink-white, brown-white, or red-brown-white and exfoliates... Dormant: Alternate twigs lack a true terminal bud and carry small ovoid imbricate buds; short shoots,...	Tolerance: Generally shade intolerant to intermediate; seedlings and crowns perform best in... Site: Favors cool, moist, well-drained, often rocky or talus-derived upland soils and is poorly suited to...

Taxon	High-value visible evidence	Silvics anchor
Gray birch *Betula populifolia*	Foliage: Alternate glossy blades are triangular to rhombic-triangular with a long narrow tip, coarse double teeth, and... Bark: Young bark is brown to reddish; saplings and small trees become chalky gray-white with dark triangular chevrons... Dormant: Slender alternate twigs lack a true terminal bud, show lenticels and small resin glands, and...	Tolerance: Very shade intolerant; recruitment and persistence require open conditions. Site: Grows from moist well-drained margins to dry infertile sands, gravels, rocky slopes, and acidic...
Sweet birch *Betula lenta*	Foliage: Alternate ovate to narrow-ovate leaves are finely and regularly double-serrate, usually with at least six teeth... Bark: Young bark is smooth reddish brown with horizontal lenticels and can resemble cherry; mature bark becomes dark... Dormant: Slender reddish-brown twigs lack a true terminal bud and carry small pointed imbricate buds; a...	Tolerance: Classed as shade intolerant overall, although seedlings benefit from side shade or... Site: Best on moist well-drained soils and protected north- or east-facing slopes; tolerates rocky shallow...
Pin cherry *Prunus pensylvanica*	Foliage: Alternate thin leaves are narrow-ovate to lanceolate, long-pointed, finely serrate, and largely glabrous; one or... Bark: Young bark is shiny reddish brown with strong horizontal lenticels; older thin bark becomes gray-brown and may... Dormant: Slender reddish-brown twigs show horizontal lenticels and small sharp imbricate buds, including a...	Tolerance: Very shade intolerant; establishment and survival are concentrated in large openings. Site: Occupies coarse to medium soils from rocky ledges and sands to moist loams, but is generally absent...
Chokecherry *Prunus virginiana*	Foliage: Alternate oblong to obovate leaves have sharply pointed teeth, mostly 8-11 pairs of lateral veins, and a mostly... Bark: Young stems are reddish to gray-brown with horizontal lenticels; older small trunks remain relatively thin and... Dormant: Slender brown to reddish twigs have obvious lenticels and sharp imbricate buds. The winter twig...	Tolerance: Shade tolerant enough to persist under forest canopy, but reaches greatest density... Site: Broadly adaptable from mesic forest edges and riparian terraces to dry-mesic thickets, provided...
American hornbeam *Carpinus caroliniana*	Foliage: Alternate elliptic to ovate leaves have a symmetrical rounded or shallowly cordate base, strong straight... Bark: Young and mature bark remains thin, smooth, blue-gray to slate gray, while the trunk develops distinctive sinewy... Dormant: Slender alternate twigs lack a true terminal bud; narrow ovoid buds have many overlapping scales...	Tolerance: Very shade tolerant as a seedling and sapling; tolerance declines with age, so... Site: Best on rich loams with high water-holding capacity, from somewhat poorly drained to well drained;...
Pagoda dogwood *Cornus alternifolia*	Foliage: Leaves are alternate, though crowded near shoot tips, with entire to slightly wavy margins and several strong... Bark: Young stems are smooth reddish to green-brown; mature bark becomes gray-brown with shallow ridges or small... Dormant: Leaves and buds are alternate, often clustered near upturned branch tips; slender twigs are...	Tolerance: Shade tolerant and able to dominate locally in mature-forest understories, while... Site: Favors cool, rich, moist, well-drained forest soils and edges; drought and heat stress increase decline.
Smooth serviceberry *Amelanchier laevis*	Foliage: Alternate ovate to elliptic leaves are sharply toothed; at flowering they are already partly expanded and... Bark: Young stems and twigs are reddish; older bark remains thin and smooth gray with pale vertical stripes, later... Dormant: Slender alternate reddish-brown twigs have narrow pointed imbricate buds, a terminal bud,...	Tolerance: Tolerates understory shade but flowers, fruits, and develops best with partial sun... Site: Prefers moist, moderately to well-drained acidic to neutral soils and tolerates rocky forest sites if...
Balsam fir *Abies balsamea*	Foliage: Needles are single, soft, flattened, aromatic when crushed, and usually arranged in two lateral ranks on lower... Bark: Sapling bark is thin, smooth, gray, and commonly dotted with conspicuous raised resin blisters. Intermediate... Dormant: Slender gray-brown twigs are relatively smooth after needles fall, with round flush leaf scars...	Tolerance: Very shade tolerant; seedlings can persist for decades as advance regeneration and... Site: Occupies moist cool sites from poorly drained organic and fine-textured lowlands to well-drained...
White spruce *Picea glauca*	Foliage: Needles are single, stiff, sharp, four-angled to somewhat flattened, and roll between the fingers; each stands... Bark: Young bark is thin, smoothish, gray to brown, becoming scaly early. Intermediate and mature bark forms thin... Dormant: Current branchlets are light brown to pink-brown, usually hairless, and coated with a thin...	Tolerance: Intermediate in shade tolerance; more tolerant than jack or red pine and the... Site: Grows across a wide moisture range but performs best on moist, well-drained, relatively fertile loams...
Black spruce *Picea mariana*	Foliage: Short single needles are four-angled, stiff, dull gray- to blue-green, glaucous, and borne around the twig on... Bark: Saplings have thin gray-brown bark with fine scales; intermediate and mature stems become dark gray to reddish... Dormant: Young yellow-brown branchlets lack bloom and are minutely hairy; at least some hairs are gland-...	Tolerance: Shade tolerant, especially as a seedling, but generally less tolerant than balsam... Site: Dominant on cold acidic peatlands, poorly drained organic soils, and nutrient-poor mineral sites,...
Red spruce *Picea rubens*	Foliage: Single sharp four-angled needles stand around the twig on woody pegs and roll between the fingers. They are... Bark: Young bark is smoothish gray-brown to reddish brown and soon develops thin flaky scales. Mature bark becomes... Dormant: Yellow-brown current branchlets are minutely hairy without a glaucous bloom; the hairs are...	Tolerance: Very shade tolerant, broadly comparable to balsam fir and more tolerant than white... Site: Best in cool humid climates on moist acidic, commonly shallow or stony soils with thick organic...
Norway spruce *Picea abies*	Foliage: Needles are single, stiff, sharp, four-angled, dark green, and borne on woody pegs around the shoot. They... Bark: Young bark is smoothish gray-brown to reddish brown, becoming finely scaly. Older trunks develop gray to brown... Dormant: Reddish-brown branchlets carry prominent woody needle pegs and pointed reddish-brown, usually...	Tolerance: Moderately shade tolerant as a young tree, but plantation growth and crown... Site: Prefers cool, moist, well-drained, commonly acidic loams or sandy loams; prolonged saturation, heat,...
Eastern white pine *Pinus strobus*	Foliage: Soft, slender, flexible blue-green needles occur in fascicles of five, one for each letter in 'white'; they are... Bark: Seedlings and young saplings have smooth thin gray-green bark. Intermediate stems become gray with shallow... Dormant: Twigs are slender and gray-brown, with a conspicuous terminal cluster of long narrow pointed...	Tolerance: Intermediate in shade tolerance; seedlings tolerate substantial shelter, but... Site: Occurs on many soils and is competitive on sandy, rocky, or glacial deposits; best growth is on moist,...
Red pine *Pinus resinosa*	Foliage: Needles occur in fascicles of two, are dark green, relatively straight, and usually 9-16 cm long. They are... Bark: Young stems have thin gray to reddish scaly bark. Intermediate and mature boles develop conspicuous orange-red... Dormant: Stout orange-brown branchlets end in clustered ovoid red-brown buds, usually resinous. Persistent...	Tolerance: Shade intolerant; seedlings and saplings need nearly full light, and crowns recede... Site: Best known on well-drained sands, gravels, and rocky ridges, though highest productivity occurs on...
Jack pine *Pinus banksiana*	Foliage: Needles occur in fascicles of two, are short, stiff, yellow- to gray-green, and usually 2-5 cm long. Each pair... Bark: Sapling bark is thin, dull gray to reddish brown, and finely scaly. Mature stems remain relatively thin-barked... Dormant: Slender gray-brown twigs bear small ovoid resinous buds and persistent paired fascicle sheaths....	Tolerance: Very shade intolerant, among the least tolerant northern conifers; seedlings... Site: Most competitive on dry nutrient-poor sands, dunes, outwash, and barrens, but also occurs on rocky...
Scots pine *Pinus sylvestris*	Foliage: Needles occur in fascicles of two, are stiff, noticeably twisted, often blue- or gray-green, and usually 3-7 cm... Bark: Young branches and upper trunks develop thin flaky cinnamon- to bright orange-brown bark, a major field clue.... Dormant: Gray- to orange-brown twigs bear clustered ovoid pointed gray-brown to reddish buds that may...	Tolerance: Shade intolerant; regeneration and good crown growth require open sun. Site: Adaptable to nutrient-poor, acidic, sandy, and seasonally dry soils if drainage is good; it performs...
Tamarack *Larix laricina*	Foliage: Soft, slender, blue- to bright-green needles occur singly on elongating current shoots and in dense tufts of... Bark: Young bark is smoothish gray to reddish brown and soon becomes finely scaly. Mature bark sheds as small thin... Dormant: Winter crowns are completely leafless but retain numerous rounded buds on short dark knobby spur...	Tolerance: Very shade intolerant; seedlings may endure light shade briefly but must reach the... Site: Characteristic of bogs, fens, and cold swamps on organic soils, yet best growth often occurs on moist,...
European larch *Larix decidua*	Foliage: Soft bright-green deciduous needles occur singly on long shoots and in dense tufts on knobby short shoots,... Bark: Young branch bark is smooth gray to yellow-brown. Mature trunks develop thick reddish- to gray-brown, deeply... Dormant: Leafless winter twigs carry many rounded buds on stout knobby spur shoots; lateral branchlets...	Tolerance: Very shade intolerant; best stem and crown development require full sun, though... Site: Prefers cool sites with moist, well-drained, moderately fertile loams or gravelly soils; waterlogging,...

Taxon	High-value visible evidence	Silvics anchor
Northern white-cedar *Thuja occidentalis*	Foliage: Tiny opposite scale leaves overlap in four ranks and completely hide the twig, forming strongly flattened fan-... Bark: Sapling bark is thin, smoothish, reddish to gray-brown, and begins peeling in narrow fibers. Mature and old bark... Dormant: Evergreen flattened sprays persist, with no conspicuous scaled winter bud; growth ends in tightly...	Tolerance: Very shade tolerant, able to persist beneath mixed canopies, although successful... Site: Occupies a broad moisture range from swamps and fens to dry cliffs, commonly favored by calcareous or...
Eastern redcedar *Juniperus virginiana*	Foliage: Adult branchlets carry tiny opposite scale leaves tightly appressed in four ranks, forming rounded to four-... Bark: Young bark is thin reddish brown to gray and begins peeling early. Mature bark shreds in long narrow fibrous... Dormant: Evergreen scale leaves or prickly juvenile leaves persist, and there is no large conspicuous...	Tolerance: Shade intolerant to only weakly tolerant; seedlings establish in open ground and... Site: Extremely site-flexible on dry rocky, sandy, clayey, and calcareous soils, including shallow...
Quaking aspen *Populus tremuloides*	Foliage: Alternate, simple leaves on normal short shoots are usually nearly round to broadly ovate or rhombic, 2-8 cm... Bark: Young stems have thin, smooth, pale greenish-white to cream bark, often with dark branch scars and horizontal... Dormant: Alternate slender brown to gray-red twigs end in narrow, pointed, usually glabrous buds with...	Tolerance: Very shade intolerant; seedlings and root suckers require nearly full light, and... Site: Occupies a very broad edaphic range, from dry sands and shallow uplands to moist loams and cold...
Bigtooth aspen *Populus grandidentata*	Foliage: Alternate, simple blades on normal shoots are broadly ovate, usually 7-12 cm long, and bear roughly 5-12... Bark: Saplings have smooth thin olive-green to gray bark with lenticels and dark branch scars. Pole bark becomes dull... Dormant: Alternate gray-brown twigs have a true terminal bud and appressed lateral buds. The several...	Tolerance: Very shade intolerant; successful suckers and seedlings need nearly full overhead... Site: Most characteristic on well-drained sandy or loamy uplands and glacial deposits, but best growth...
Balsam poplar *Populus balsamifera*	Foliage: Alternate, simple blades are broadly ovate to lance-ovate with a long acute or short-acuminate tip, a broad-... Bark: Young stems have smooth gray-green to brown bark with evident lenticels; pole bark remains relatively smooth and... Dormant: The large, sharply pointed terminal bud is conspicuously resinous, sticky, and strongly balsamic-...	Tolerance: Very shade intolerant; regeneration needs nearly full light and is rapidly... Site: Best developed on moist, nutrient-rich, aerated alluvial soils and fresh flood deposits; it also...
Speckled alder *Alnus incana subsp. rugosa*	Foliage: Alternate, simple leaves are narrow-ovate to elliptic or broadly elliptic, commonly with a rounded to subcordate... Bark: Young brown to gray stems are smooth and densely speckled with conspicuous horizontal or oval white lenticels,... Dormant: Alternate ovoid buds stand on obvious 2-4 mm stalks and are covered by two, sometimes three,...	Tolerance: Generally shade intolerant to intermediate; dense thickets persist where crowns... Site: Characteristic of saturated or seasonally flooded mineral and organic soils along streams, shorelines,...
American mountain-ash *Sorbus americana*	Foliage: Leaves are alternate and pinnately compound, generally 10-22 cm long, with a terminal leaflet and numerous... Bark: Young trunks are thin, smooth, gray to gray-brown, with horizontal lenticels; older small boles become somewhat... Dormant: Alternate twigs carry a conspicuous ovoid to pointed terminal bud and smaller lateral buds with...	Tolerance: Shade intolerant overall; it can persist beneath a semiopen canopy but abundance,... Site: Most successful on cool moist acidic soils with adequate aeration, but occurs from swamp borders to...
Showy mountain-ash *Sorbus decora*	Foliage: Alternate pinnately compound leaves commonly carry 13-17 blue-green leaflets. Compared with American mountain-... Bark: Young gray-brown bark is smooth with lenticels; older small boles become scaly or shallowly fissured but remain... Dormant: Alternate twigs end in a conspicuous ovoid to pointed scaly bud. The margins of the winter bud...	Tolerance: Tolerates partial shade but not a deeply closed canopy; crown growth, flowering,... Site: Favored by cool moist conditions and occurs on a variety of acidic to neutral substrates, including...
Silver maple *Acer saccharinum*	Foliage: Leaves are opposite, simple, and usually five-lobed, with very deep sinuses that produce long narrow lobes; the... Bark: Seedlings and saplings have smooth light-gray bark, sometimes with fine vertical lenticels. Intermediate stems... Dormant: Twigs are opposite, slender to moderately stout, reddish brown, and bear blunt red to reddish-...	Tolerance: Intermediate to moderately tolerant when young, but best crown development and... Site: Most characteristic on moist alluvial soils and seasonally flooded lowlands; it tolerates considerable...
Boxelder *Acer negundo*	Foliage: Leaves are opposite and pinnately compound, an unusual combination among regional trees. Each leaf usually has... Bark: Young bark is smooth, thin, and gray to light brown. Intermediate stems develop shallow narrow fissures and low... Dormant: Opposite stout twigs are green, olive, purplish, or brown and often carry a pale waxy bloom. Buds...	Tolerance: Shade intolerant to intermediate when young; it persists best along edges and in... Site: Most frequent on moist fertile alluvium, stream margins, and low terraces, but remarkably tolerant of...
River birch *Betula nigra*	Foliage: Leaves are alternate, simple, and broadly rhombic to ovate, usually with a wedge-shaped base, acute tip, and... Bark: Young stems and upper branches have salmon, cinnamon, pinkish tan, or cream bark peeling in thin papery curls... Dormant: Slender alternate twigs are reddish brown to gray, lenticellate, and may bear glandular hairs...	Tolerance: Shade intolerant; seedlings need open light and established crowns deteriorate... Site: Best on moist, well-aerated alluvial sands, silts, and loams along waterways; it tolerates periodic...
White ash *Fraxinus americana*	Foliage: Leaves are opposite and pinnately compound, usually with seven but sometimes five to nine stalked leaflets.... Bark: Seedlings and young saplings have smooth gray bark with scattered pale lenticels. Pole-size bark develops... Dormant: Stout opposite brown to blue-brown twigs are normally glabrous. The terminal bud is rounded at...	Tolerance: Young seedlings tolerate shade for a time, but tolerance declines with age;... Site: Best on deep, fertile, moist but well-drained loams with favorable nutrient status; it avoids...
Green ash *Fraxinus pennsylvanica*	Foliage: Leaves are opposite and pinnately compound with five to nine stalked leaflets. Leaflets are usually serrate and... Bark: Young bark is smooth gray-brown with lenticels. Pole and mature stems develop narrow interlacing ridges and... Dormant: Opposite gray-brown branchlets are usually pubescent on current growth, and the terminal bud is...	Tolerance: Variable from intermediate to moderately tolerant when young, with advance... Site: Most characteristic on fertile alluvial clays, silts, and loams subject to periodic flooding, yet...
Black ash *Fraxinus nigra*	Foliage: Leaves are opposite and pinnately compound with seven to thirteen, occasionally more, narrow sharply serrate... Bark: Sapling bark is smooth light gray and may feel somewhat corky. Intermediate stems develop shallow fissures and... Dormant: Stout opposite twigs bear dark brown to nearly black buds. The terminal bud stands distinctly...	Tolerance: Intermediate when young, with seedlings able to persist under a partial canopy but... Site: Specialized for cold, poorly drained mineral or organic soils, often where nutrient-rich groundwater...
Eastern cottonwood *Populus deltoides*	Foliage: Leaves are alternate, simple, broadly triangular or deltoid, and often nearly as wide as long, with a truncate... Bark: Seedlings and young branches are smooth yellow-green to gray with lenticels. Intermediate trunks become gray and... Dormant: Twigs are stout, glossy yellow-brown to reddish brown, conspicuously lenticellate, and often...	Tolerance: Very shade intolerant; seedlings and lower crowns decline rapidly when overtopped. Site: Best on deep moist, well-aerated sands and silts deposited along rivers; tolerates periodic flooding...
American elm *Ulmus americana*	Foliage: Leaves are alternate, simple, usually seven to fourteen centimeters long, ovate to elliptic, abruptly pointed,... Bark: Young bark is smooth gray to gray-brown. Intermediate stems form shallow intersecting fissures and flattened... Dormant: Slender alternate twigs form a slight zigzag and generally lack corky wings. Small ovoid brown...	Tolerance: Intermediate to moderately tolerant when young; seedlings can persist beneath a... Site: Broad site amplitude from moist well-drained uplands to seasonally flooded rich alluvium, with best...
Slippery elm *Ulmus rubra*	Foliage: Leaves are alternate, simple, usually eight to sixteen centimeters long, broad ovate to elliptic, double-... Bark: Young bark is gray-brown and shallowly fissured. Intermediate and mature bark becomes dark gray to reddish brown... Dormant: Alternate zigzag twigs are more often stout and hairy than those of American elm. Ovoid buds have...	Tolerance: Intermediate to tolerant when young, allowing seedlings to persist beneath partial... Site: Best on moist fertile, well-drained loams and calcareous rich woods; occurs on riparian terraces and...

Taxon	High-value visible evidence	Silvics anchor
Rock elm *Ulmus thomasii*	Foliage: Leaves are alternate, simple, elliptic to obovate or ovate, double-serrate, and usually firm textured, with a... Bark: Young stems are smooth gray-brown. Intermediate bark develops shallow fissures and narrow firm ridges. Mature... Dormant: Alternate twigs are gray, brown, or reddish and lack a true terminal bud. Two- to several-year...	Tolerance: Intermediate in shade tolerance; seedlings persist under partial canopy but need... Site: Most characteristic on well-drained loams, rocky uplands, and calcareous substrates, though it also...
Siberian elm *Ulmus pumila*	Foliage: Leaves are alternate, simple, small, usually two to six and a half centimeters long, elliptic to lanceolate, and... Bark: Young bark is smooth gray to light brown. Intermediate stems develop shallow fissures and narrow ridges. Mature... Dormant: Slender alternate gray-brown twigs form a zigzag and lack a true terminal bud. Small globose buds...	Tolerance: Shade intolerant; seedlings establish and grow best in open fields, roadsides,... Site: Extremely tolerant of drought, cold, wind, alkaline or nutrient-poor soils, and compaction, while also...
Black willow *Salix nigra*	Foliage: Leaves are alternate, simple, narrowly lanceolate, mostly about 7.5-17 millimeters wide, long-pointed, and... Bark: Young bark is smooth yellow-brown, gray, or reddish brown. Intermediate stems become shallowly fissured with... Dormant: Slender alternate branchlets are yellow, orange, red-brown, or gray, conspicuously lenticellate,...	Tolerance: Very shade intolerant; seedlings and lower crowns decline rapidly beneath taller... Site: Requires a continuous growing-season water supply and is most competitive on very wet alluvial silts,...
American sycamore *Platanus occidentalis*	Foliage: Leaves are alternate, simple, broad, palmately veined, and usually three to five shallow or moderate lobes with... Bark: Young stems are gray-brown and smoothish. Intermediate trunks begin exfoliating in irregular thin plates that... Dormant: Stout alternate twigs zigzag between nodes and show a complete stipule-scar ring encircling each...	Tolerance: Very shade intolerant; seedlings need open light and crowns self-prune or decline... Site: Best on deep moist, fertile, well-aerated alluvial loams and sands along streams; tolerates periodic...
Northern red oak *Quercus rubra*	Foliage: Alternate leaves usually have 7-11 bristle-tipped lobes, sinuses generally extending less than halfway to the... Bark: Young stems are smooth gray-brown. Saplings and pole timber develop long smooth pale vertical bands between... Dormant: Stout reddish-brown to gray-brown twigs end in a cluster of several pointed, reddish-brown,...	Tolerance: Intermediate in shade tolerance; seedlings can persist for a time beneath partial... Site: Best on deep, fertile, moist but well-drained loams and silt loams; it occupies a broad site range but...
Black oak *Quercus velutina*	Foliage: Alternate leaves are extremely variable, usually with 5-9 bristle-tipped lobes and sinuses ranging from shallow... Bark: Young bark is smooth gray. It darkens early and becomes thick, deeply furrowed, and blocky on mature stems,... Dormant: Twigs are moderately stout and reddish-brown to gray, ending in a cluster of relatively large,...	Tolerance: Intermediate but light-demanding for successful recruitment; seedlings may persist... Site: Occurs from dry sandy and rocky uplands to moist rich well-drained loams; it avoids prolonged...
Scarlet oak *Quercus coccinea*	Foliage: Alternate glossy leaves usually have 7-9 narrow bristle-tipped lobes and deep broad C- or U-shaped sinuses... Bark: Sapling bark is smooth gray-brown. Mature bark becomes dark gray to nearly black with irregular narrow ridges... Dormant: Slender reddish-brown twigs end in clustered ovoid, pointed reddish-brown buds that may be finely...	Tolerance: Intermediate to intolerant and strongly light-demanding for successful... Site: Most characteristic of dry, acidic, well-drained sandy or rocky soils on ridges and upper slopes;...
Northern pin oak *Quercus ellipsoidalis*	Foliage: Alternate leaves commonly have 5-7 narrow bristle-tipped lobes separated by deep rounded sinuses, with a mostly... Bark: Young bark is smooth gray-brown; older stems develop shallow fissures and irregular dark gray to blackish... Dormant: Reddish-brown twigs end in a clustered group of small ovoid to conic reddish-brown buds, hairless...	Tolerance: Very intolerant of shade; it does not regenerate successfully beneath its own... Site: Specialized on dry acidic, nutrient-poor sands and sandy or sandstone uplands with a thin organic...
Pin oak *Quercus palustris*	Foliage: Alternate leaves usually have 5-7 narrow bristle-tipped lobes divided by deep broad U-shaped sinuses. The lower... Bark: Young and pole-size trees retain thin, smooth gray-brown bark for many years. Older boles become shallowly... Dormant: Slender reddish-brown twigs end in clusters of small pointed chestnut-brown buds that are usually...	Tolerance: Intolerant of shade; overtopped seedlings and intermediates decline quickly, and... Site: Characteristic of poorly drained, seasonally flooded heavy clays, glacial till flats, and bottomlands;...
White oak *Quercus alba*	Foliage: Alternate leaves have 7-9 rounded, bristleless lobes, with sinuses ranging from shallow to deep and an often... Bark: Young stems are smooth light gray to brownish gray. Pole and mature bark becomes pale ashy gray and breaks into... Dormant: Gray to reddish twigs terminate in a tight cluster of short, blunt to ovoid reddish-brown, mostly...	Tolerance: Intermediate and generally more shade tolerant as a seedling than the most light-... Site: Occupies a broad range from dry ridges to moist well-drained coves, growing best on deep fertile loams...
Swamp white oak *Quercus bicolor*	Foliage: Alternate obovate leaves have shallow irregular rounded lobes or coarse rounded teeth and a wedge-shaped base.... Bark: Young bark is smooth gray-brown. Older bark becomes gray, scaly, and irregularly flaky, often peeling in strips... Dormant: Yellow-brown to gray-brown twigs carry clustered short blunt buds; young twigs may be somewhat...	Tolerance: Intermediate; seedlings can establish under moderate shade but require release for... Site: Most common on hydromorphic mineral, alluvial, or organic soils with imperfect to poor drainage and...
Bur oak *Quercus macrocarpa*	Foliage: Alternate leaves are large and obovate with rounded bristleless lobes, commonly narrowed by a pair of very deep... Bark: Young bark is gray and becomes corky at an early age. Mature boles have very thick dark gray-brown bark divided... Dormant: Stout gray-brown twigs end in clustered short rounded buds and commonly develop conspicuous corky...	Tolerance: Intermediate but unable to recruit indefinitely under dense shade; seedlings and... Site: Exceptionally broad, from droughty sands, thin limestone ridges, claypans, and prairie loams to moist...
Chestnut oak *Quercus montana*	Foliage: Alternate obovate to elliptic leaves are unlobed but have 10-16 pairs of large rounded crenate teeth, resembling... Bark: Young bark is smooth gray-brown. Mature bark becomes exceptionally thick, dark gray-brown, and deeply divided by... Dormant: Stout reddish- to gray-brown twigs end in clustered pointed reddish-brown buds, sometimes finely...	Tolerance: Intermediate; seedlings may persist beneath an open canopy but successful... Site: Most characteristic of dry, rocky, well-drained acidic sandstone and shale ridges with low fertility,...
Common hackberry *Celtis occidentalis*	Foliage: Alternate simple ovate leaves have a strongly unequal, oblique base, three prominent veins arising near the... Bark: Young bark is smooth gray-brown with lenticels. Intermediate stems develop isolated corky warts, and mature gray... Dormant: Slender alternate twigs are often zigzag, reddish-brown to gray, with small appressed ovoid...	Tolerance: Intermediate to tolerant; seedlings may persist in dense shade, although prolonged... Site: Grows on many soils but is strongest on fertile valley alluvium, riparian terraces, and calcareous or...
American chestnut *Castanea dentata*	Foliage: Leaves are alternate, simple, narrowly elliptic to lanceolate, commonly 12-22 cm long, with a long pointed tip... Bark: Seedlings, sprouts, and young stems have smooth olive-brown to gray-brown bark with conspicuous pale lenticels.... Dormant: Alternate slender reddish-brown twigs are glabrous and bear small ovoid to conical brown buds...	Tolerance: Seedlings and sprouts can persist for years under partial shade, but sustained... Site: Historically occupied a wide range but was especially successful on well-drained acidic loams and...
Bitternut hickory *Carya cordiformis*	Foliage: Leaves are alternate and pinnately compound, usually with 7 or 9 narrowly ovate to lanceolate serrate leaflets;... Bark: Saplings have relatively smooth gray bark. Pole and mature stems develop shallow furrows and narrow interlacing... Dormant: The high-value character is a slender pointed terminal bud with sulfur-yellow to mustard-yellow,...	Tolerance: Intermediate; seedlings can persist under partial canopy, but vigorous height... Site: Best on deep, fertile, moist, well-aerated loams of bottomlands, terraces, and rich uplands, though it...
Shagbark hickory *Carya ovata*	Foliage: Alternate pinnately compound leaves usually have 5 leaflets, less often 7; the terminal leaflet and upper... Bark: Seedlings and saplings have smooth gray bark, and pole trees develop close shallow ridges before any shag... Dormant: Twigs are stout, alternate, and contain solid pith. The terminal bud is large, ovoid, and brown...	Tolerance: Intermediate; seedlings can survive beneath partial canopy but require release to... Site: Grows from dry uplands to moist lowlands, with best growth on deep fertile well-drained loams; it is...

Taxon	High-value visible evidence	Silvics anchor
Pignut hickory *Carya glabra*	Foliage: Alternate pinnately compound leaves most often have 5 leaflets, sometimes 7; the upper three are larger, and... Bark: Young stems are smooth gray. Mature bark is gray to dark gray with close, interlacing, often diamond-patterned... Dormant: Alternate slender to moderately stout twigs have solid pith and a small ovoid to ellipsoid...	Tolerance: Intermediate overall but often treated as relatively intolerant in the Northeast;... Site: Most characteristic of well-drained dry to mesic uplands, ridges, and slopes on loams or rocky soils;...
Mockernut hickory *Carya tomentosa*	Foliage: Alternate pinnately compound leaves usually have 7 leaflets, sometimes 5 or 9. Leaflets are broad, serrate, and... Bark: Sapling bark is gray and relatively smooth. Mature bark becomes dark gray with deep furrows and close... Dormant: Twigs are notably stout and often densely pubescent, with solid pith and large three-lobed leaf...	Tolerance: Intermediate; established seedlings can persist beneath partial cover, but good... Site: Most characteristic of well-drained dry to mesic uplands on loams and rocky soils, while best growth...
Black walnut *Juglans nigra*	Foliage: Alternate pinnately compound leaves commonly have 11-23 finely serrate leaflets. The terminal leaflet is absent,... Bark: Young bark is gray-brown and relatively smooth with lenticels. Pole bark develops narrow fissures, while mature... Dormant: Alternate stout brown twigs show large shield- or heart-shaped leaf scars with three main groups...	Tolerance: Shade intolerant; seedlings may survive briefly under partial cover but need early... Site: Best on deep, fertile, moist, well-drained loams with near-neutral reaction, often on alluvium,...
Butternut *Juglans cinerea*	Foliage: Alternate pinnately compound leaves usually have 11-17 serrate leaflets and a well-developed terminal leaflet.... Bark: Young bark is smooth gray with lenticels. Mature bark is light gray to ashy, divided by shallow furrows into... Dormant: Stout gray-brown current twigs are densely hairy, often with a hairy ridge or pad immediately...	Tolerance: Shade intolerant; seedlings and sprouts need substantial light, and established... Site: Best on fertile moist well-drained soils of stream terraces, coves, and slopes, often with calcareous...
Blackgum *Nyssa sylvatica*	Foliage: Alternate simple leaves are elliptic to obovate, usually entire but occasionally with a few coarse teeth, with a... Bark: Young trunks are gray and shallowly furrowed. Mature bark develops irregular vertical ridges and then breaks... Dormant: Alternate gray-brown to reddish twigs bear small ovoid to conical, reddish-brown, many-scaled...	Tolerance: Tolerant; seedlings can persist under closed hardwood cover, although near-full... Site: Extremely broad site amplitude, from dry acidic ridges to moist loams and stream bottoms; best growth...
Sassafras *Sassafras albidum*	Foliage: Alternate simple leaves can be unlobed and oval, asymmetrically one-lobed like a mitten, or symmetrically three-... Bark: Young stems and branches are smooth green to reddish green with pale lenticels. Pole bark turns orange-brown and... Dormant: Alternate greenish to reddish aromatic twigs have solid pith and a true terminal bud. Vegetative...	Tolerance: Shade intolerant; seedlings and root sprouts grow best in full or nearly full... Site: Occupies many soils but is most conspicuous on well-drained sandy or loamy old fields, ridges, and...
Cucumbertree *Magnolia acuminata*	Foliage: Alternate simple leaves are large, commonly 10-25 cm long, elliptic to oblong, entire, and abruptly to gradually... Bark: Young stems have thin smooth gray-brown bark with scattered lenticels. Pole bark begins to fissure; mature bark... Dormant: Alternate stout reddish-brown twigs bear large silky gray-green to brown terminal buds enclosed...	Tolerance: Intermediate; young plants tolerate some side shade, but regeneration is more... Site: Best on deep, fertile, moist, well-drained loams of coves, valleys, and lower slopes; poorly drained,...
Tuliptree *Liriodendron tulipifera*	Foliage: Alternate simple leaves are broadly four-lobed, with two lobes on each side and a squared, truncate, or... Bark: Seedlings and saplings have smooth dark greenish-gray bark with pale vertical lenticels. Pole bark develops... Dormant: Alternate twigs are reddish brown, aromatic when lawfully scraped or broken, and ringed at each...	Tolerance: Shade intolerant; seedlings may persist briefly under light cover, but dominant... Site: Best on deep, fertile, moist, well-drained loams of coves, lower slopes, and alluvial terraces; growth...
Flowering dogwood *Cornus florida*	Foliage: Leaves and twigs are opposite. Simple oval to ovate blades have entire or slightly wavy margins and three to... Bark: Young stems are smooth gray to gray-brown. As stems mature, bark breaks into many small, nearly square or... Dormant: Opposite slender twigs end either in a narrow pointed vegetative bud or a conspicuous rounded,...	Tolerance: Tolerant of understory shade, though flowering, fruiting, and crown vigor improve... Site: Occurs from dry uplands to moist stream-adjacent soils but is most frequent on well-drained,...
Pitch pine *Pinus rigida*	Foliage: Stout, rigid, yellow-green needles are usually in fascicles of three, less often two, with persistent basal... Bark: Young bark is reddish-brown and scaly; intermediate and mature boles develop thick, dark gray-brown irregular... Dormant: Stout shoots carry clustered resinous pointed buds, persistent fascicle sheaths, and often short...	Tolerance: Shade intolerant; seedlings and sprouts need strong light, and associated... Site: Most characteristic on infertile acidic sands, gravels, and shallow rocky soils, but also occupies...
Atlantic white-cedar *Chamaecyparis thyoides*	Foliage: Minute opposite scale leaves overlap in four ranks on slender blue-green branchlets that appear more four-angled... Bark: Young bark is thin reddish-brown; older stems separate into narrow flat fibrous strips and shallow intersecting... Dormant: Evergreen sprays retain tiny appressed scales through winter; buds are inconspicuous and...	Tolerance: Generally shade intolerant; seedlings tolerate brief partial cover but successful... Site: Restricted mainly to saturated, acidic, low-nutrient organic soils, peat, muck, and wet sands where...
Black locust *Robinia pseudoacacia*	Foliage: Alternate pinnately compound leaves usually carry 7 to 19 oval, thin, entire leaflets with a terminal leaflet.... Bark: Young stems are smooth greenish to reddish-brown with lenticels; older bark becomes dark gray-brown and deeply... Dormant: Zigzag alternate twigs have small buds recessed beneath the leaf scar and often a pair of short...	Tolerance: Very shade intolerant; rapid juvenile growth occurs in full light, while shaded... Site: Adaptable to many well-drained soils, including droughty, eroded, compacted, and nutrient-poor...
Tree-of-heaven *Ailanthus altissima*	Foliage: Very large alternate pinnately compound leaves carry many long leaflets that are otherwise nearly entire but... Bark: Sapling bark is smooth greenish-gray to pale gray with conspicuous lenticels; older bark remains relatively... Dormant: Stout tan to reddish twigs have very large heart- or shield-shaped leaf scars with many bundle...	Tolerance: Seedlings tolerate some partial shade briefly, but the species is fundamentally... Site: Extremely adaptable to dry, compacted, polluted, rocky, calcareous, and disturbed soils, though growth...
White mulberry *Morus alba*	Foliage: Alternate simple leaves on one shoot may be unlobed, mitten-shaped, or deeply multi-lobed; margins are serrate,... Bark: Young bark is smooth yellowish- to orange-brown with lenticels; mature bark becomes gray-brown to orange-brown... Dormant: Slender alternate zigzag twigs end in a false terminal bud and carry small appressed ovoid buds...	Tolerance: Best in full sun and tolerant of partial shade, but generally unable to recruit or... Site: Adaptable to moist or dry well-drained soils, urban compaction, varied pH, and disturbed ground;...

Part III - Fillable instructional sessions

Complete sessions in order when possible. Closed-book retrieval means close or hide the teaching atlas before answering.

Session A - Rapid identification and remediation triage

Purpose: rapid unfamiliar drills with no new taxa. Taxon scope: all 80 taxa prioritized by certification deficit. Feedback: after batch submission.

Before the session: which species or comparison currently feels least reliable?

Exit retrieval: state the most important correction or strengthened distinction

☐ Session A complete (enter X)

Session B - Cumulative scientific-name retrieval

Purpose: closed-response nomenclature batches. Taxon scope: all 80 taxa balanced in both name directions. Feedback: after batch submission.

Before the session: which species or comparison currently feels least reliable?

1. Silvics

A stand prescription lists Black walnut as an objective species. State the site conditions that favor it and one site or climatic constraint that would make that objective questionable.

Response closed-book retrieval

2. Silvics

For Balsam fir (Abies balsamea), describe the moisture, drainage, and soil setting a forester should expect, then place it in its climatic or forest-association context.

Response closed-book retrieval

3. Forest Association

A field crew reports Northern white-cedar from an unfamiliar stand. State the regional and community context that would make the report credible, then request morphological evidence needed to confirm it.

Response closed-book retrieval

4. Silvics

For Heart-leaved paper birch (Betula cordifolia), describe the moisture, drainage, and soil setting a forester should expect, then place it in its climatic or forest-association context.

Response closed-book retrieval

5. Silvics

Give one operational management implication for Atlantic white-cedar and explain its response to an appropriate silvicultural system or cutting pattern.

Response closed-book retrieval

6. Silvics

After a harvest, what regeneration pathway should a forester expect from Bitternut hickory, and what substrate, light, moisture, or advance-regeneration conditions control success?

Response closed-book retrieval

7. Silvics

A stand containing Balsam poplar is disturbed. Explain the likely biological response and one important damaging agent or vulnerability that changes the prognosis.

Response closed-book retrieval

8. Forest Association

A field crew reports River birch from an unfamiliar stand. State the regional and community context that would make the report credible, then request morphological evidence needed to confirm it.

Response closed-book retrieval

9. Silvics

Predict the role of Gray birch from suppression through canopy recruitment. Tie the prediction to shade tolerance, successional status, and growth or longevity.

Response closed-book retrieval

10. Nomenclature

Give the preferred course common name for Magnolia acuminata.

Response closed-book retrieval

Exit retrieval: state the most important correction or strengthened distinction

☐ Session B complete (enter X)

Session C - Confusers and insufficient evidence

Purpose: isolated and side-by-side confuser drills with requested-view reasoning. Taxon scope: all 24 confusion sets plus uncertainty items. Feedback: after response.

Before the session: which species or comparison currently feels least reliable?

Sugar / black / Norway maple

Course separator: Petiole latex; leaf underside hair and stipules; bud size/shape; samara angle

My visible separator

Best resolving view

Red / silver maple

Course separator: Sinus depth; lower-surface whiteness; spring samara size; twig odor; bark strips

My visible separator

Best resolving view

Striped / mountain maple

Course separator: Green-white striped bark versus unstriped; leaf size/base; fruit cluster posture; growth form

My visible separator

Best resolving view

White / green / black ash

Course separator: Leaf-scar shape and bud position; leaflet stalks; samara wing extent; black-ash corky bark

My visible separator

Best resolving view

Paper / heart-leaved / gray birch

Course separator: Leaf base and vein count; twig resin glands; peeling versus tight bark; black branch marks; elevation

My visible separator

Best resolving view

Yellow / sweet birch

Course separator: Both wintergreen; yellow birch golden curling bark and spur shoots versus sweet birch dark blocky bark

My visible separator

Best resolving view

White / black / red / Norway spruce

Course separator: Cone size and scale margin; twig hairs; bud resin; foliage color/odor; pendulous branchlets

My visible separator

Best resolving view

Spruce / fir / hemlock

Course separator: Woody pegs versus pad base versus petiole; needle cross-section; cone orientation and persistence

My visible separator

Best resolving view

Red / jack / Scots / pitch pine

Course separator: Fascicle count/length; bend test; cone curvature/prickles; upper-bole orange bark; epicormic shoots

My visible separator

Best resolving view

Tamarack / European larch

Course separator: Cone size and scale count/edge; needle cluster density; native peatland versus planting

My visible separator

Best resolving view

Red-oak group

Course separator: Bud hair/size; lower-leaf hair; sinus architecture; acorn cup depth/rings; bark; site and branch angle

My visible separator

Best resolving view

White-oak group

Course separator: Lower-leaf felt; fruit stalk; fringed cup; chestnut teeth; bark architecture; site

My visible separator

Best resolving view

Aspens / balsam poplar / cottonwood

Course separator: Petiole flattening; tooth size; leaf base; bud size/stickiness/odor; bark and floodplain/upland context

My visible separator

Best resolving view

American / slippery / rock / Siberian elm

Course separator: Leaf roughness/base; bud hair; corky wings; samara hair/notch; bark layering; crown

My visible separator

Best resolving view

Black / pin / chokecherry

Course separator: Fruit cluster architecture; petiole glands; lower-midvein hair; bark age; whole pioneer/clonal form

My visible separator

Best resolving view

American hornbeam / eastern hophornbeam

Course separator: Muscular smooth bole and three-lobed fruit bracts versus shreddy bark and hoplike sacs

My visible separator

Best resolving view

Black walnut / butternut

Course separator: Terminal leaflet; rachis/twig hair; light versus dark pith; round versus elongated nut; bark

My visible separator

Best resolving view

Bitternut / shagbark / pignut / mockernut hickory

Course separator: Terminal bud color/size/hair; leaflet count; rachis hair; husk thickness/dehiscence; bark

My visible separator

Best resolving view

Northern white-cedar / Atlantic white-cedar / eastern redcedar

Course separator: Flattened fans versus fine false-cypress sprays versus rounded juniper twigs; cone type; fruit color; wetland chemistry

My visible separator

Best resolving view

Native compound leaves / invasive confusers

Course separator: Opposite/alternate; bud position; leaflet glands; thorns; pith; leaf scar; odor; fruit type

My visible separator

Best resolving view

Yellow birch / black cherry bark

Course separator: Golden horizontal curls and wintergreen twig versus burnt-chip plates, petiole glands, and almond twig odor

My visible separator

Best resolving view

Serviceberry / mountain-ashes / chokecherry

Course separator: Simple versus compound leaves; pome calyx versus drupe; leaflet proportions; buds; fruit architecture

My visible separator

Best resolving view

Flowering / pagoda dogwood

Course separator: Opposite versus alternate branching; red versus blue-black fruit; flower bud; crown tiers

My visible separator

Best resolving view

White mulberry / sassafras / basswood

Course separator: Latex and variable coarse leaves; sassafras aroma/mixed shapes; basswood asymmetric heart base and bracted fruit

My visible separator

Best resolving view

Exit retrieval: state the most important correction or strengthened distinction

☐ Session C complete (enter X)

Session D - Certification modality rehearsal

Purpose: separate foliage bark reproductive dormant and whole-tree stations. Taxon scope: all 80 taxa blueprint balanced by tier and modality. Feedback: after station batch submission.

Before the session: which species or comparison currently feels least reliable?

Practice 1

Identity: common OR scientific name

Evidence asset INAT_PHOTO_718752564 | credit in Sources appendix

Only the identity field is universally required. Instructors may score reasoning fields where the rubric
Use genus or 'insufficient evidence' when the photograph cannot defend species.

Required - defensible identity

Confidence: low / moderate / high

Nearest plausible alternative

Visible separator / evidence

Best additional view if unresolved

Practice 2

Identity: common OR scientific name

Evidence asset INAT_PHOTO_708163540 | credit in Sources appendix

Only the identity field is universally required. Instructors may score reasoning fields where the rubric
Use genus or 'insufficient evidence' when the photograph cannot defend species.

Required - defensible identity

Confidence: low / moderate / high

Nearest plausible alternative

Visible separator / evidence

Best additional view if unresolved

Practice 3

Identity: common OR scientific name

Evidence asset INAT_PHOTO_718818603 | credit in Sources appendix

Only the identity field is universally required. Instructors may score reasoning fields where the rubric
Use genus or 'insufficient evidence' when the photograph cannot defend species.

Required - defensible identity

Confidence: low / moderate / high

Nearest plausible alternative

Visible separator / evidence

Best additional view if unresolved

Practice 4

Identity: common OR scientific name

Evidence asset INAT_PHOTO_599227949 | credit in Sources appendix

Only the identity field is universally required. Instructors may score reasoning fields where the rubric
Use genus or 'insufficient evidence' when the photograph cannot defend species.

Required - defensible identity

Confidence: low / moderate / high

Nearest plausible alternative

Visible separator / evidence

Best additional view if unresolved

Practice 5

Identity: common OR scientific name

Evidence asset INAT_PHOTO_675502402 | credit in Sources appendix

Only the identity field is universally required. Instructors may score reasoning fields where the rubric
Use genus or 'insufficient evidence' when the photograph cannot defend species.

Required - defensible identity

Confidence: low / moderate / high

Nearest plausible alternative

Visible separator / evidence

Best additional view if unresolved

Practice 6

Identity: common OR scientific name

Evidence asset INAT_PHOTO_725671642 | credit in Sources appendix

Only the identity field is universally required. Instructors may score reasoning fields where the rubric

Use genus or 'insufficient evidence' when the photograph cannot defend species.

Required - defensible identity

Confidence: low / moderate / high

Nearest plausible alternative

Visible separator / evidence

Best additional view if unresolved

Practice 7

Identity: common OR scientific name

Evidence asset INAT_PHOTO_456409461 | credit in Sources appendix

Only the identity field is universally required. Instructors may score reasoning fields where the rubric

Use genus or 'insufficient evidence' when the photograph cannot defend species.

Required - defensible identity

Confidence: low / moderate / high

Nearest plausible alternative

Visible separator / evidence

Best additional view if unresolved

Practice 8

Identity: common OR scientific name

Evidence asset INAT_PHOTO_682970601 | credit in Sources appendix

Only the identity field is universally required. Instructors may score reasoning fields where the rubric
Use genus or 'insufficient evidence' when the photograph cannot defend species.

Required - defensible identity

Confidence: low / moderate / high

Nearest plausible alternative

Visible separator / evidence

Best additional view if unresolved

Practice 9

Identity: common OR scientific name

Evidence asset INAT_PHOTO_705662705 | credit in Sources appendix

Only the identity field is universally required. Instructors may score reasoning fields where the rubric
Use genus or 'insufficient evidence' when the photograph cannot defend species.

Required - defensible identity

Confidence: low / moderate / high

Nearest plausible alternative

Visible separator / evidence

Best additional view if unresolved

Practice 10

Identity: common OR scientific name

Evidence asset INAT_PHOTO_707712143 | credit in Sources appendix

Only the identity field is universally required. Instructors may score reasoning fields where the rubric
Use genus or 'insufficient evidence' when the photograph cannot defend species.

Required - defensible identity

Confidence: low / moderate / high

Nearest plausible alternative

Visible separator / evidence

Best additional view if unresolved

Exit retrieval: state the most important correction or strengthened distinction

☐ Session D complete (enter X)

Session E - Silvics and regional synthesis

Purpose: oral or written synthesis and final targeted retrieval. Taxon scope: all 80 taxa prioritized by silvics and field condition deficits. Feedback: after batch submission.

Before the session: which species or comparison currently feels least reliable?

1. Nomenclature

Translate this preferred common name into the canonical binomial: Jack pine.

Response closed-book retrieval

2. Family Relationship

To which botanical family does Pignut hickory (Carya glabra) belong?

Response closed-book retrieval

3. Nomenclature

Translate this preferred common name into the canonical binomial: Pin cherry.

Response closed-book retrieval

4. Nomenclature

Give the preferred course common name for Magnolia acuminata.

Response closed-book retrieval

5. Silvics

Design the species-specific part of a prescription for Black cherry: state what to protect or control, and justify the cutting or regeneration approach from its known response.

Response closed-book retrieval

6. Silvics

A stand prescription lists Flowering dogwood as an objective species. State the site conditions that favor it and one site or climatic constraint that would make that objective questionable.

Response closed-book retrieval

7. Silvics

For White ash (Fraxinus americana), describe the moisture, drainage, and soil setting a forester should expect, then place it in its climatic or forest-association context.

Response closed-book retrieval

8. Forest Association

A field crew reports Flowering dogwood from an unfamiliar stand. State the regional and community context that would make the report credible, then request morphological evidence needed to confirm it.

Response closed-book retrieval

9. Silvics

Predict how Norway spruce responds to disturbance or release, and identify one browse, insect, disease, fire, flooding, or injury concern a field forester should carry into the stand.

Response closed-book retrieval

10. Silvics

Place Silver maple in succession: give its shade tolerance, explain how it recruits or persists, and characterize its longevity or growth strategy.

Response closed-book retrieval

Exit retrieval: state the most important correction or strengthened distinction

☐ Session E complete (enter X)

Image credits and instructional references

Image credits apply only to the labeled teaching and formative practice plates in this public learner packet. Secure examination specimens and answer-bearing metadata are not included.

Asset	Creator / license	Source
INAT_PHOTO_456409461	(c) Loïc Mathieu, some rights reserved (CC BY) | cc-by	https://www.inaturalist.org/photos/456409461
INAT_PHOTO_599227949	(c) Steven Lamonde, some rights reserved (CC BY) | cc-by	https://www.inaturalist.org/photos/599227949
INAT_PHOTO_675502402	(c) JSReiff, some rights reserved (CC BY) | cc-by	https://www.inaturalist.org/photos/675502402
INAT_PHOTO_682970601	(c) Sam Turner, some rights reserved (CC BY) | cc-by	https://www.inaturalist.org/photos/682970601
INAT_PHOTO_705662705	(c) Alan Prather, some rights reserved (CC BY) | cc-by	https://www.inaturalist.org/photos/705662705
INAT_PHOTO_707712143	(c) Alan Prather, some rights reserved (CC BY) | cc-by	https://www.inaturalist.org/photos/707712143
INAT_PHOTO_708163540	no rights reserved | cc0	https://www.inaturalist.org/photos/708163540
INAT_PHOTO_718752564	(c) Quinten Wiegersma, some rights reserved (CC BY) | cc-by	https://www.inaturalist.org/photos/718752564
INAT_PHOTO_718818603	no rights reserved | cc0	https://www.inaturalist.org/photos/718818603
INAT_PHOTO_725671642	(c) Cheryl Stinchcomb, some rights reserved (CC BY) | cc-by	https://www.inaturalist.org/photos/725671642